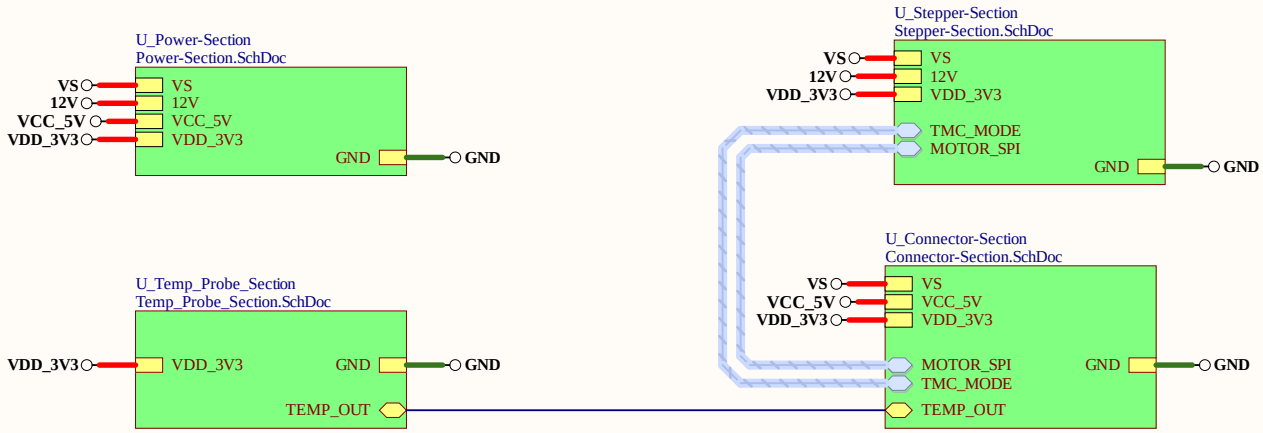
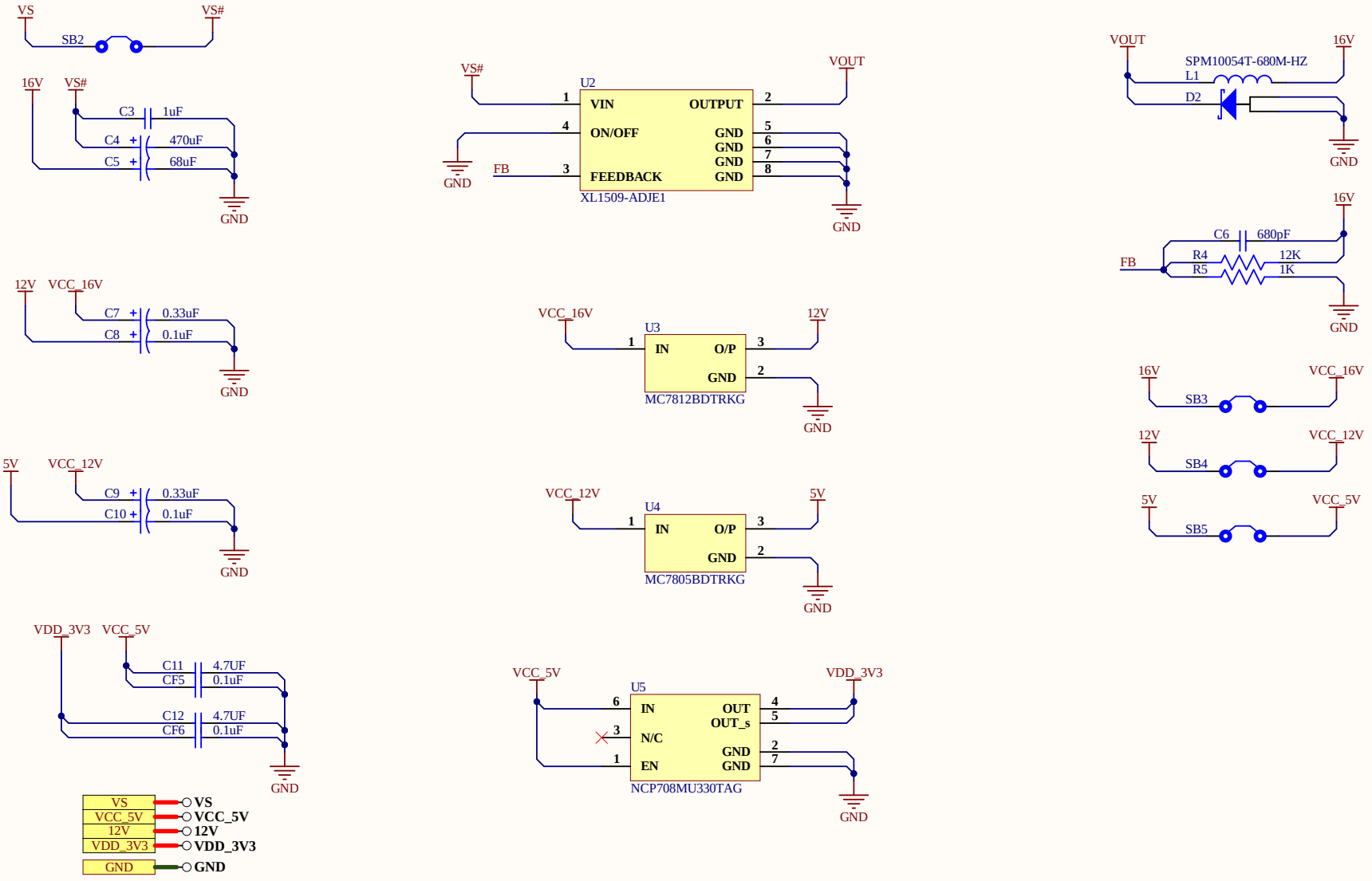
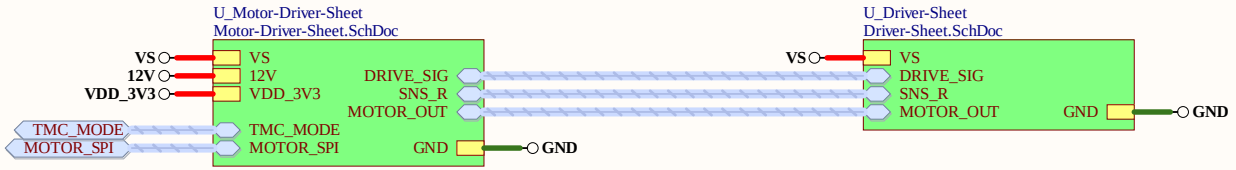
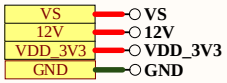
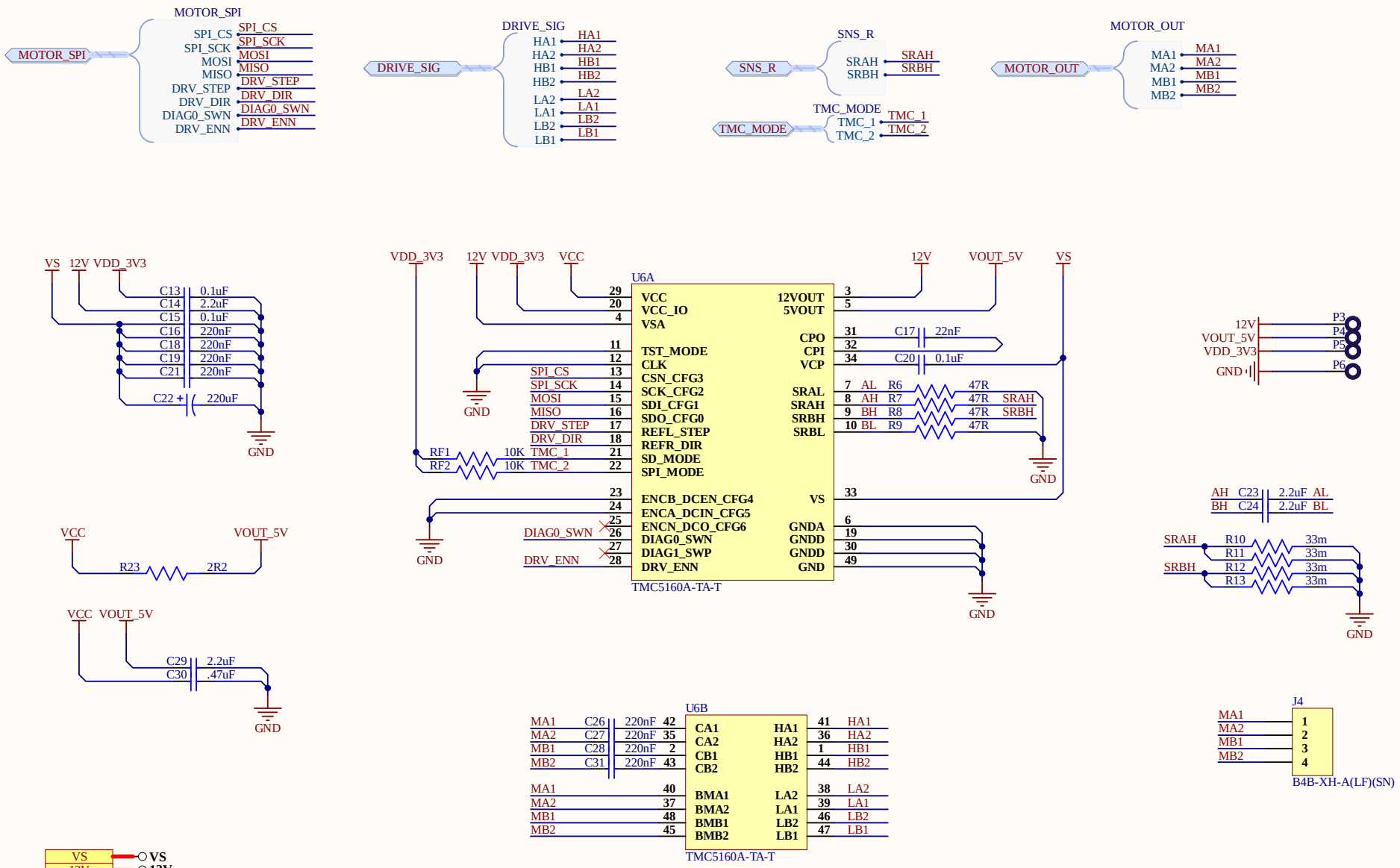
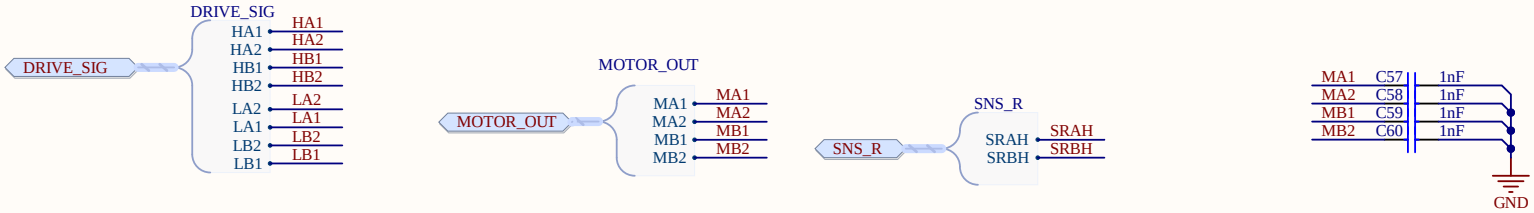

U_Power-Section
Power-Section.SchDoc
VS
12V
VCC_5V
VDD_3V3
VS
12V
VCC_5V
VDD_3V3
GND
GND
U_Stepper-Section
Stepper-Section.SchDoc
VS
12V
VDD_3V3
VS
12V
VDD_3V3
TMC_MODE
MOTOR_SPI
GND
GND
U_Temp_Probe_Section
Temp_Probe_Section.SchDoc
VDD_3V3
VDD_3V3
GND
GND
TEMP_OUT
U_Connector-Section
Connector-Section.SchDoc
VS
VCC_5V
VDD_3V3
VS
VCC_5V
VDD_3V3
MOTOR_SPI
TMC_MODE
GND
GND
TEMP_OUT

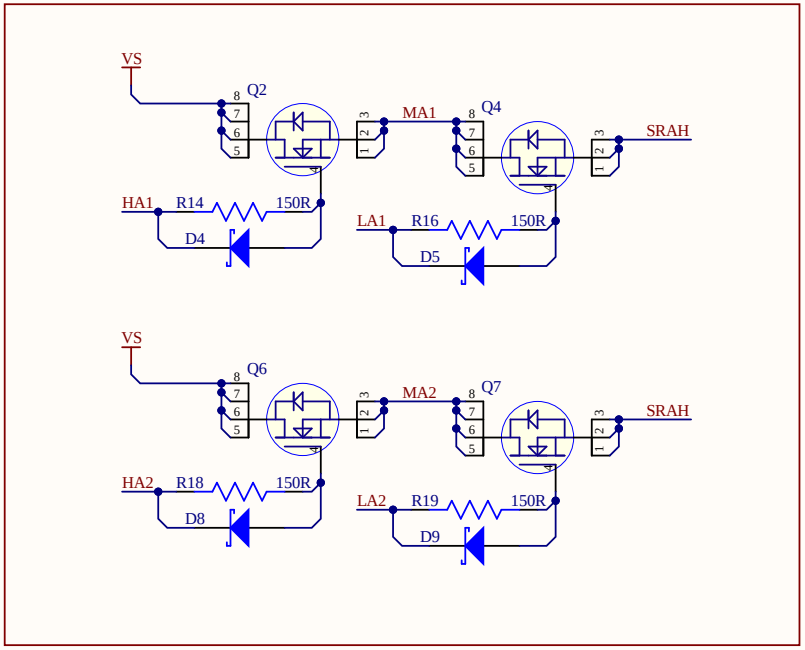

VS
VS#
SB2
16V
C3 1uF
C4 470uF
C5 68uF
GND
12V
VCC_16V
C7 0.33uF
C8 0.1uF
5V
VCC_12V
C9 0.33uF
C10 0.1uF
VDD_3V3
VCC_5V
C11 4.7UF
CF5 0.1uF
C12 4.7UF
CF6 0.1uF
VS
VCC_5V
12V
VDD_3V3
GND
U2
VIN
ON/OFF
FEEDBACK
OUTPUT
GND
FB
XL1509-ADJE1
U3
IN
O/P
MC7812BDTRKG
U4
MC7805BDTRKG
U5
N/C
EN
OUT
OUT_s
NCP708MU330TAG
VOUT
SPM10054T-680M-HZ
L1
D2
C6 680pF
R4 12K
R5 1K
SB3
SB4
SB5

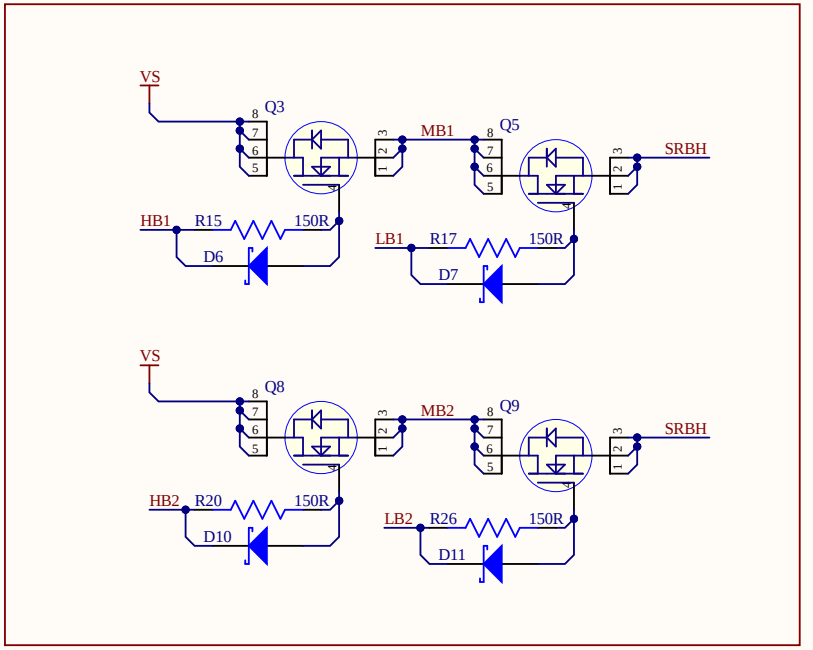

U_Motor-Driver-Sheet
Motor-Driver-Sheet.SchDoc
VS
12V
VDD_3V3
TMC_MODE
MOTOR_SPI
VS
12V
VDD_3V3
TMC_MODE
MOTOR_SPI
DRIVE_SIG
SNS_R
MOTOR_OUT
GND
GND
U_Driver-Sheet
Driver-Sheet.SchDoc
VS
VS
DRIVE_SIG
SNS_R
MOTOR_OUT
GND
GND


VS
12V
VDD_3V3
GND
VS
12V
VDD_3V3
GND


MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
DRIVE_SIG
HA1
HA2
HB1
HB2
LA2
LA1
LB2
LB1
SNS_R
SRAH
SRBH
TMC_MODE
TMC_1
TMC_2
MOTOR_OUT
MA1
MA2
MB1
MB2
VS
12V
VDD_3V3
GND
C13 0.1uF
C14 2.2uF
C15 0.1uF
C16 220nF
C18 220nF
C19 220nF
C21 220nF
C22 220uF
VCC
VOUT_5V
R23 2R2
C29 2.2uF
C30 .47uF
U6A
TMC5160A-TA-T
RF1 10K
RF2 10K
C17 22nF
C20 0.1uF
R6 47R
R7 47R
R8 47R
R9 47R
U6B
C26 220nF
C27 220nF
C28 220nF
C31 220nF
J4
B4B-XH-A(LF)(SN)
P3
P4
P5
P6
AH C23 2.2uF AL
BH C24 2.2uF BL
R10 33m
R11 33m
R12 33m
R13 33m


DRIVE_SIG
HA1
HA2
HB1
HB2
LA2
LA1
LB2
LB1
MOTOR_OUT
MA1
MA2
MB1
MB2
SNS_R
SRAH
SRBH
C57 1nF
C58 1nF
C59 1nF
C60 1nF
GND

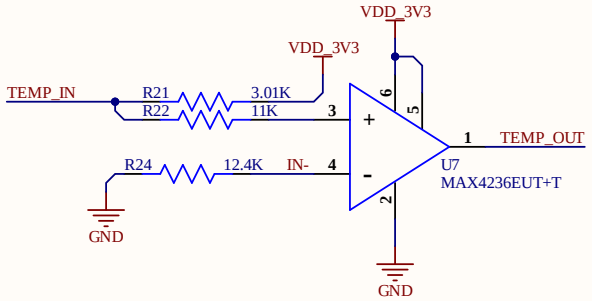

VS
Q2
MA1
Q4
SRAH
HA1
R14
150R
D4
LA1
R16
150R
D5
VS
Q6
MA2
Q7
SRAH
HA2
R18
150R
D8
LA2
R19
150R
D9

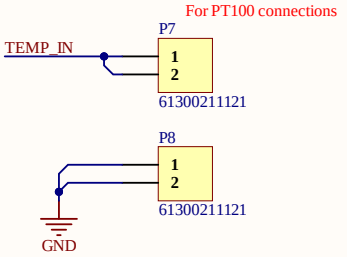

VS
Q3
MB1
Q5
SRBH
HB1
R15
150R
D6
LB1
R17
150R
D7
VS
Q8
MB2
Q9
SRBH
HB2
R20
150R
D10
LB2
R26
150R
D11

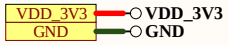

VS
GND
VS
GND

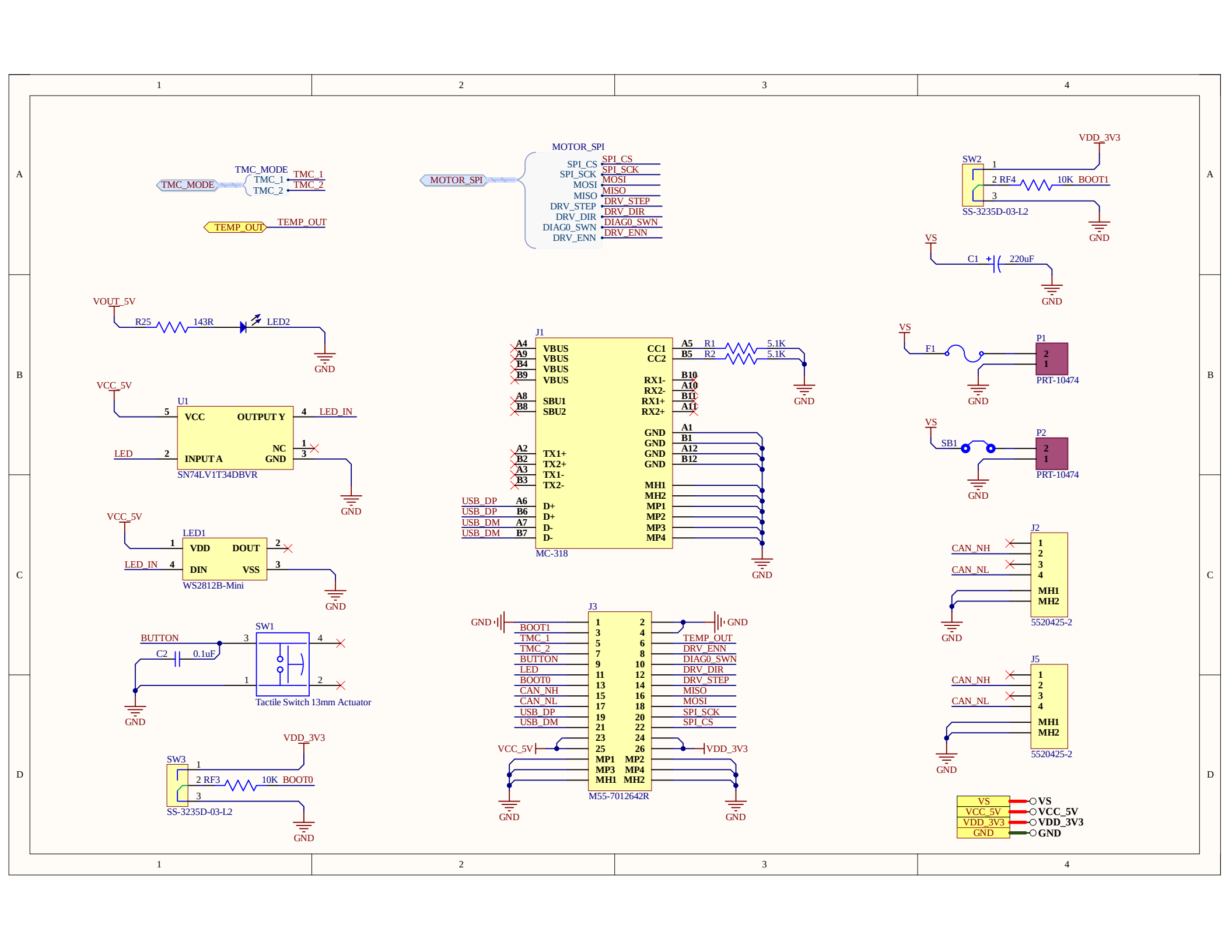

TEMP_OUT
TEMP_OUT


TEMP_OUT
R27
11.8K
TEMP_IN
R28
105K
IN-


VDD_3V3
VDD_3V3
TEMP_IN
R21
3.01K
R22
11K
3
6
5
+
1
TEMP_OUT
R24
12.4K
IN-
4
-
U7
MAX4236EUT+T
2
GND
GND


For PT100 connections
TEMP_IN
P7
1
2
61300211121
P8
1
2
61300211121
GND


VDD_3V3
GND
VDD_3V3
GND


TMC_MODE
TMC_1
TMC_2
TEMP_OUT
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
VDD_3V3
SW2
2 RF4
10K BOOT1
SS-3235D-03-L2
GND
VS
C1
220uF
VOUT_5V
R25
143R
LED2
VCC_5V
U1
VCC
OUTPUT Y
LED_IN
INPUT A
NC
SN74LV1T34DBVR
LED
J1
VBUS
SBU1
SBU2
TX1+
TX2+
TX1-
TX2-
D+
D-
CC1
CC2
RX1-
RX2-
RX1+
RX2+
GND
MC-318
R1
R2
5.1K
USB_DP
USB_DM
F1
P1
PRT-10474
SB1
P2
LED1
VDD
DOUT
DIN
VSS
WS2812B-Mini
J2
CAN_NH
CAN_NL
MH1
MH2
5520425-2
J5
SW1
BUTTON
C2
0.1uF
Tactile Switch 13mm Actuator
J3
BOOT1
TMC_1
TMC_2
BUTTON
LED
BOOT0
CAN_NH
CAN_NL
USB_DP
USB_DM
TEMP_OUT
DRV_ENN
DIAG0_SWN
DRV_DIR
DRV_STEP
MISO
MOSI
SPI_SCK
SPI_CS
MP1
MP2
MP3
MP4
M55-7012642R
SW3
2 RF3
10K BOOT0
SS-3235D-03-L2
VS
VCC_5V
VDD_3V3
GND